

Introduction

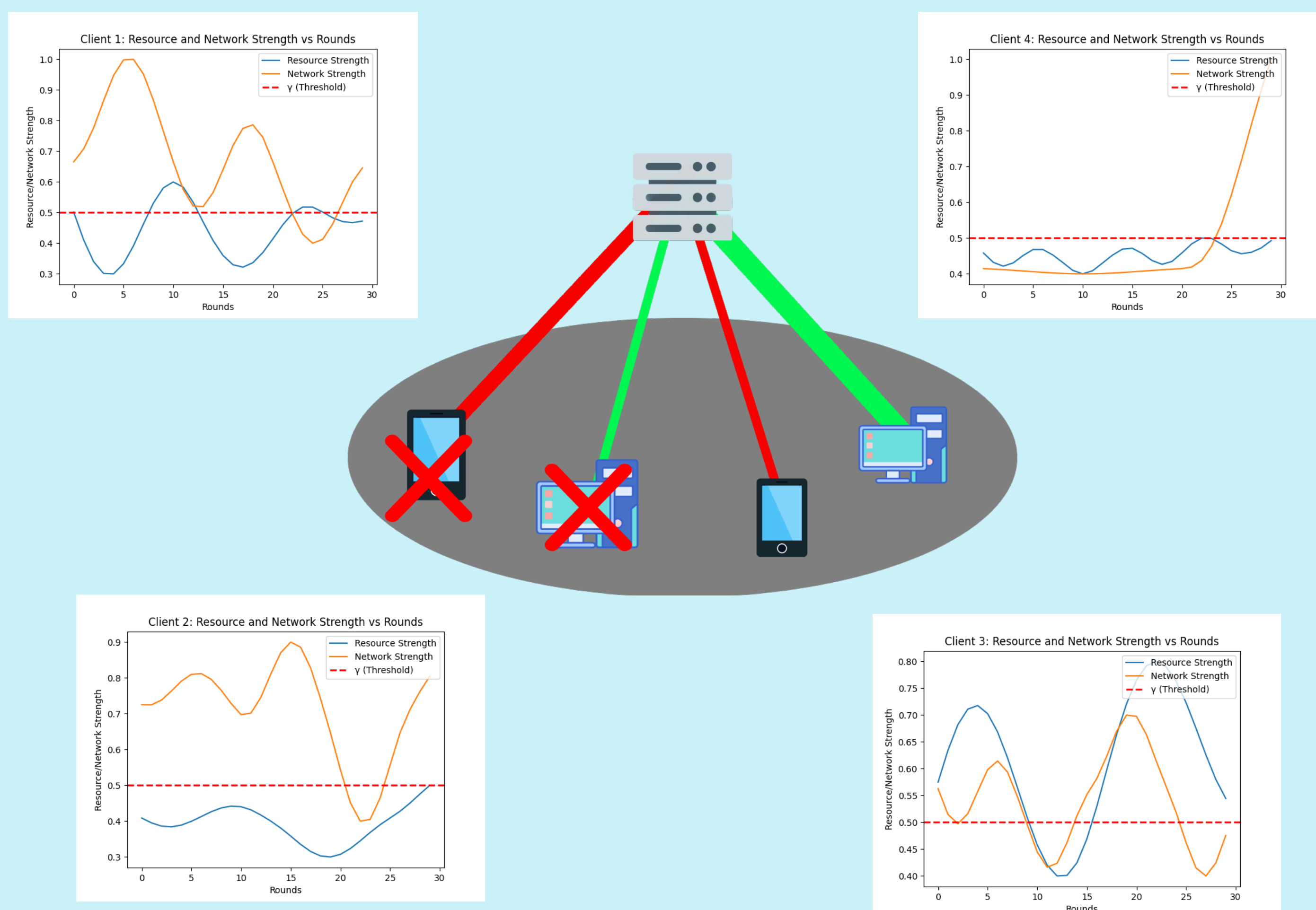
Federated Learning (FL) in mobile edge and IoT environments is frequently hindered by extreme data heterogeneity and inherent client unreliability. While most research focuses on "stragglers" (slow devices), ADaRES addresses the critical problem of complete client unavailability, where data from entire classes may become inaccessible.

- The "Unavailability" Crisis: In non-IID settings, if clients holding specific classes go offline (due to low battery or high CPU), those data modalities are permanently lost from the training pool.
- The ADaRES Objective: To create a resilient framework that redistributes data from resource-poor but network-accessible clients to stable peers, ensuring no data modalities are lost.
- Privacy Model: Operates within a trusted edge domain where peer-to-peer encrypted replicas are permitted, but server-side data exposure is strictly avoided.

Methodology

The ADaRES framework manages data availability and computational load at the start of each Federated Learning round. We use Perlin noise to simulate realistic, time-dependent resource fluctuations in edge devices.

- Dynamic State Detection: Clients are assessed against a global threshold ' Γ ' for Resource (R_i) and Network (N_i) strength. We target "Down" clients who can offload data but cannot train.
- Similarity-Aware Reallocation: Data from offline clients is moved to the Top-K stable hosts. A similarity factor ' α ' ensures data is sent to peers with statistically similar datasets.
- Adaptive Epoch Scheduling: To balance the total system effort, clients with smaller, original datasets have their local epoch counts (E_h) decreased. This ensures that host peers with expanded data loads have the relative priority and time needed to fully integrate reallocated information.



Algorithm 1 ADaRES: Redistribute Data

Require: t (round), Γ (threshold), α (sim. factor), K_{top} , L_{max} (max load)

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1:  $C_{down} \leftarrow \{i \mid R_i(t) < \Gamma \text{ and } N_i(t) \geq \Gamma\}$ 
2:  $C_{up} \leftarrow \{j \mid R_j(t) \geq \Gamma \text{ and } N_j(t) \geq \Gamma\}$ 
3: for  $d \in C_{down}$  (not already reassigned) do
4:    $Scores \leftarrow []$ 
5:   for  $h \in C_{up}$  (where  $Load(h) < L_{max}$ ) do
6:     if  $\alpha < 0$  then
7:        $S_h \leftarrow -\alpha \cdot Sim(d, h) + (1 + \alpha) \cdot R_h(t)$ 
8:     else
9:        $S_h \leftarrow \alpha \cdot Sim(d, h) + (1 - \alpha) \cdot R_h(t)$ 
10:    end if
11:     $Scores.append(h, S_h)$ 
12:  end for
13:   $ChosenHosts \leftarrow TopK(Scores, K_{top})$ 
14:  Split and assign  $Data_d$  to  $ChosenHosts$ 
15:  Record assignment in  $TempAssignments[d]$ 
16: end for
17: for  $h \in C_{up}$  do
18:    $E_h \leftarrow E_{base} - f(Load(h), E_{min})$  {Reduce epochs}
19: end for

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Algorithm 2 ADaRES: Recover Reassignments

Require: t (round), Γ (threshold)

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1:  $C_{recovered} \leftarrow \{i \mid R_i(t) \geq \Gamma \text{ and } i \in TempAssignments\}$ 
2: Reset all  $E_i$  to  $E_{base}$ 
3: for  $d \in C_{recovered}$  do
4:   Retrieve data from hosts listed in  $TempAssignments[d]$ 
5:   Restore data to  $d$ 
6:   Remove  $d$  from  $TempAssignments$ 
7: end for

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Algorithm 3 ADaRES: Full Training Round

Require: Rounds, Γ , α

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1: for  $t \in \{1, \dots, Rounds\}$  do
2:   Call Recover Reassignments( $t, \Gamma$ )
3:   Call Redistribute Data( $t, \Gamma, \alpha, \dots$ )
4:    $Updates \leftarrow []$ 
5:   for  $i \in \{1, \dots, N\}$  (where  $R_i(t) \geq \Gamma$ ) do
6:      $w_i \leftarrow ClientTrain(w_{global}, Data_i, E_i)$ 
7:     if  $N_i(t) \geq \Gamma$  then
8:        $Updates.append(w_i)$ 
9:     end if
10:  end for
11:   $w_{global} \leftarrow Aggregate(Updates)$ 
12:  for  $i \in \{1, \dots, N\}$  (where  $N_i(t) \geq \Gamma$ ) do
13:    SendModel( $i, w_{global}$ )
14:  end for
15: end for

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Results

Experimental validation on the CIFAR-10 dataset demonstrates that ADaRES successfully overcomes the challenges of unstable edge environments.

- Final Accuracy: ADaRES achieves 76.9% accuracy, which is a 27.8% improvement over standard models and matches an ideal "always available" system.
- Stability vs. Reality: While the conventional "Reality" model fails to converge (49.1%) due to missing data classes, ADaRES exhibits excellent stability by ensuring all data modalities are represented in every round.
- Efficiency Gains: ADaRES delivers this performance using only 3.74 average epochs, compared to 3.96 in the failing model and 10.0 in the baseline, representing a massive reduction in total computational effort.
- Host Selection: Testing various ' α ' values showed that the act of reallocation is more critical than the specific choice of host, confirming that data availability is the primary driver of model convergence.

Model	Final Accuracy	Improvement vs Reality
Baseline 10	0.7689 (76.9%)	+27.8%
Baseline 8	0.6067 (60.7%)	+11.6%
ADaRES ($\alpha = -1$)	0.7692 (76.9%)	+27.8%
ADaRES ($\alpha = -0.5$)	0.7672 (76.7%)	+27.6%
ADaRES ($\alpha = 0$)	0.7677 (76.8%)	+27.7
ADaRES ($\alpha = 0.5$)	0.7662 (76.6%)	+27.5%
ADaRES ($\alpha = 1$)	0.7521 (75.2%)	+26.1%
Reality	0.4907 (49.1%)	-

Model	Avg. Epochs (per client/round)
Baseline 10	10.00
Baseline 8	8.00
Reality	3.96
ADaRES (all α)	3.74

Conclusion

ADaRES serves as a resilient and computationally efficient solution to the challenge of client unavailability in Federated Learning.

- Key Findings: Proactive data reallocation preserves data modalities and maintains high accuracy in highly unstable IoT environments.
- Constraints: The framework requires sufficient network bandwidth for peer-to-peer transfers and is designed specifically for trusted-edge deployments with flexible data mobility.
- Future Directions: We intend to incorporate AI-driven predictive models to estimate the duration of client outages, allowing for more intelligent decisions to minimize costly data transfers.

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